A Simple Characterization of the Gamma Function
Author(s): Detlef Laugwitz and Bernd Rodewald
Source: *The American Mathematical Monthly*, Vol. 94, No. 6 (Jun. - Jul., 1987), pp. 534-536
Published by: Taylor & Francis on behalf of the Mathematical Association of America
Stable URL: https://www.jstor.org/stable/2322846
Accessed: 29-03-2026 10:34 UTC

JSTOR is a not-for-profit service that helps scholars, researchers, and students discover, use, and build upon a wide range of content in a trusted digital archive. We use information technology and tools to increase productivity and facilitate new forms of scholarship. For more information about JSTOR, please contact support@jstor.org.

Your use of the JSTOR archive indicates your acceptance of the Terms & Conditions of Use, available at https://about.jstor.org/terms

Taylor & Francis, Mathematical Association of America are collaborating with JSTOR to digitize, preserve and extend access to *The American Mathematical Monthly*

This content downloaded from 147.88.241.4 on Sun, 29 Mar 2026 10:34:42 UTC
All use subject to https://about.jstor.org/terms

PROPOSITION 3. *If f is an odd function on $[a, -a]$, then p_1 is its best convex (and concave) approximation.*

REFERENCES

1. H. G. Burchard, Extremal Positive Splines with Applications to Interpolation and Approximation by Generalized Convex Functions, Bull. Amer. Math. Soc., 79 (1973), 959–964.
2. T. J. Rivlin, An Introduction to the Approximation of Functions, Dover, New York, 1981.
3. A. W. Roberts and D. E. Varberg, Convex Functions, Academic Press, New York, 1973.
4. V. A. Ubhaya, An $O(n)$ Algorithm for Discrete n-Point Convex Approximation with Applications to Continuous Case, J. Math. Anal. Appl., 72 (1979), 338–354.
5. D. V. Widder, The Laplace Transform, Princeton University Press, Princeton, 1946.

A Simple Characterization of the Gamma Function

DETLEF LAUGWITZ and BERND RODEWALD
Fachbereich Mathematik, Technische Hochschule, Darmstadt, Federal Republic of Germany

The important properties $\Gamma(n+1) = n!$ for natural numbers n and $\Gamma(x+1) = x\Gamma(x)$ for $x > 0$ do not characterize the Gamma function. A well-known additional property sufficient for a characterization is the convexity of $\log \Gamma(x)$ [**1**]. We feel that a property originally considered by Euler is less far-fetched and much easier to use.

The Gamma function made its first appearance in a letter of Euler to Goldbach in 1729 when an infinite product was announced. Later Euler produced several representations and, finally, disclosed his original idea [**2**]: For a fixed infinitely large natural number n and all finite natural numbers m, the expression $(n+m)!$ behaves approximately like a geometric sequence. In other words, $L(n+m) = \log(n+m)!$ is infinitesimally close to a linear function of m:

$$\begin{aligned} L(n+m) &= \log n! + \sum_{k=1}^{m} \log(n+k) \\ &= L(n) + m \cdot \log(n+1) + \sum_{k=1}^{m} \log\left(1 + \frac{k-1}{n+1}\right) \\ &\approx L(n) + m \cdot \log(n+1). \end{aligned}$$

Now the assumption of Euler is that this linearity property extends to all finite real numbers x in place of m. A proof in the spirit of Euler should explicitly use infinitely large and small numbers [**3**]. Presently, we shall give the theorem and its proof in terms of limits.

This content downloaded from 147.88.241.4 on Sun, 29 Mar 2026 10:34:42 UTC
All use subject to https://about.jstor.org/terms

THEOREM. *There exists a unique function* Γ, $\Gamma(x) > 0$ *for* $x \geqslant 1$, *such that* $L(x) = \log \Gamma(x+1)$ *has, for* $x \geqslant 0$, *the following properties*:

(i) $L(0) = 0$ (*initial condition*, $1! = 1$),

(ii) $L(x+1) = \log(x+1) + L(x)$ (*functional equation*),

(iii) $L(n+x) = L(n) + x \cdot \log(n+1) + r_n(x)$, *where*

$$\lim_{n \to \infty} r_n(x) = 0. \tag{1}$$

Proof. From (i) and (ii), we obtain

$$L(n) = \log n! = \sum_{k=1}^{n} \log k \tag{2}$$

and, for natural n and real $x \geqslant 0$,

$$L(x+n) = L(x) + \sum_{k=1}^{n} \log(x+k). \tag{3}$$

As a consequence of (3), (iii) and (2),

$$L(x) = L(n) - \sum_{k=1}^{n} \log(x+k) + x \cdot \log(n+1) + r_n(x), \tag{4}$$

$$L(x) = \sum_{k=1}^{n} [\log k - \log(x+k) + x(\log(k+1) - \log k)] + r_n(x), \tag{5}$$

$$L(x) = \sum_{k=1}^{n} [(1-x)\log k + x \cdot \log(k+1) - \log(x+k)] + r_n(x). \tag{6}$$

By $\log(n+1) = -\gamma_n + \sum_{k=1}^{n} \frac{1}{k}$, $\lim_{n \to \infty} \gamma_n = \gamma = .577\ldots$, we obtain from (4) and (2)

$$L(x) = -\gamma_n x + \sum_{k=1}^{n} \left(\frac{x}{k} - \log(x+k) + \log k \right) + r_n(x), \tag{7}$$

$$L(x) = -\gamma_n x + \sum_{k=1}^{n} \left(\frac{x}{k} - \log\left(1 + \frac{x}{k}\right) \right) + r_n(x). \tag{8}$$

The series of (8) converges, since for $k > x \geqslant 0$

$$0 \leqslant \frac{x}{k} - \log\left(1 + \frac{x}{k}\right) = \frac{x}{k} - \sum_{j=1}^{\infty} \frac{(-1)^{j+1}}{j} \left(\frac{x}{k}\right)^j \leqslant \frac{x^2}{2k^2}.$$

It follows that the right-hand sides of (8), and consequently of (7), (6), (5) and (4), converge, and that

$$L(x) = -\gamma x + \sum_{k=1}^{\infty} \left(\frac{x}{k} - \log\left(1 + \frac{x}{k}\right) \right)$$

is a uniquely determined real-valued function for $x \geqslant 0$ which can be represented by any of those equations for $n \to \infty$, where $r_n(x) \to 0$.

This content downloaded from 147.88.241.4 on Sun, 29 Mar 2026 10:34:42 UTC
All use subject to https://about.jstor.org/terms

It remains to show that $L(x)$ has properties (i), (ii), (iii). Here (i) follows, e.g., from (8). From (5) and (1),

$$\begin{aligned} L(x+1) - L(x) &= \sum_{k=1}^{n} [\log(x+k) - \log(x+k+1) + \log(k+1) - \log k] \\ &\quad + r_n(x+1) - r_n(x) \\ &= \log(x+1) - \log(x+n+1) + \log(n+1) \\ &\quad + r_n(x+1) - r_n(x) \\ &= \log(x+1) - \log\left(1 + \frac{x}{n+1}\right) + r_n(x+1) - r_n(x). \end{aligned}$$

For $n \to \infty$ we obtain (ii). Now (3) is a consequence of (ii), and (4) is equivalent to (8). Postulate (iii) follows from (3) and (4). □

Immediate consequences of (6) and (8) are the infinite products for $\Gamma(x+1) = x \cdot \Gamma(x) = \exp L(x)$,

$$x \cdot \Gamma(x) = \prod_{k=1}^{\infty} \frac{k^{1-x}(k+1)^x}{x+k} \quad \text{(Euler 1729)} \tag{6'}$$

and

$$x \cdot \Gamma(x) = e^{-\gamma x} \prod_{k=1}^{\infty} \frac{e^{x/k}}{1 + \frac{x}{k}}. \tag{8'}$$

It is well known how the products can be used to derive the integral representation and the properties of $\Gamma(z)$ for complex z [**4**].

REFERENCES

1. Emil Artin, The Gamma Function, Holt, Rinehart and Winston, 1964; Einführung in die Theorie der Gammafunktion, Teubner, 1931.
2. Leonhard Euler, Institutiones Calculi Differentialis, 1755; Opera omnia (1), 10.
3. Detlef Laugwitz and Bernd Rodewald, Auf Eulers Spuren zur Gamma-funktion (to appear).
4. E. T. Whittaker and G. N. Watson, A Course of Modern Analysis, reprint Cambridge University Press, 1965, pp. 241ff.

This content downloaded from 147.88.241.4 on Sun, 29 Mar 2026 10:34:42 UTC
All use subject to https://about.jstor.org/terms